How-To Library

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Run a profile on Databricks Delta tables using Azure Databricks with ODBC connection

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Abstract

Use this article to learn the steps to run a profile on Databricks Delta tables using Azure Databricks with ODBC connection.

Supported Versions

- Cloud Data Profiling April 2022

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Overview

You can run a profile on Databricks Delta tables using an ODBC driver.

You can download the Databricks ODBC driver here: https://databricks.com/spark/odbc-drivers-download.

Run a profile using Azure Databricks with ODBC connection on Windows

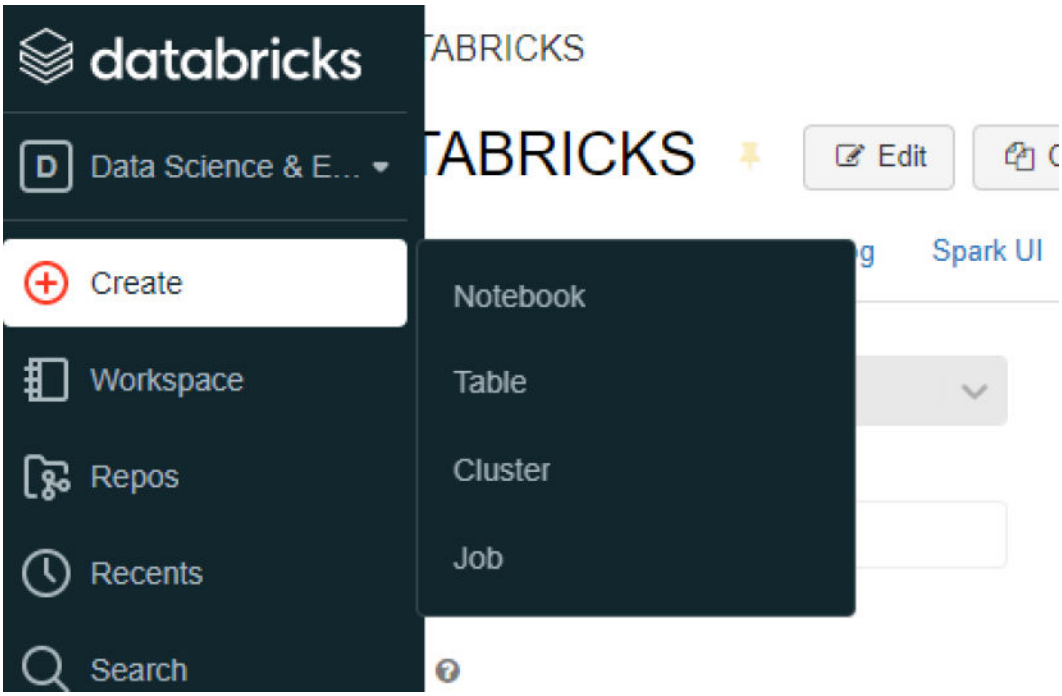
You can run a profile on Databricks Delta tables using Azure Databricks with ODBC connection on Windows.

Step 1. Create a cluster in Databricks

Create a new cluster in Databricks or use an existing cluster.

Before creating a new cluster, check for existing clusters in the **Clusters** tab of the Azure Databricks portal. If there is an existing cluster, you can restart the cluster.

1. Login to the Azure Databricks portal.
2. Go to your workspace.
3. Click **Create** and choose **Cluster** as shown in the following image:



4. Enter a name for the cluster.

   You can continue with the default values for Worker type and Driver type.

5. Click **Create Cluster** and wait for the cluster to be up.

   Note: If you are using an existing cluster, make sure that the cluster is up and running.

Step 2. Retrieve the ODBC details

Retrieve the ODBC details from the Databricks cluster.

1. Go to the **JDBC/ODBC** tab of the Databricks cluster.
2. Retrieve the ODBC details including the sever host name, port, protocol, HTTP path, and the URL.

Step 3: Install and configure the ODBC driver for Windows

Set up a Data Source Name (DSN) configuration in Windows to connect the ODBC client application to Databricks.

To set up a DSN configuration, you can use the Windows ODBC Data Source Administrator. For more information on the ODBC Data Source Administrator, see https://docs.microsoft.com/en-us/sql/database-engine/configure-windows/open-the-odbc-data-source-administrator?view=sql-server-ver15.

1. Download the latest Databricks ODBC driver for Windows.

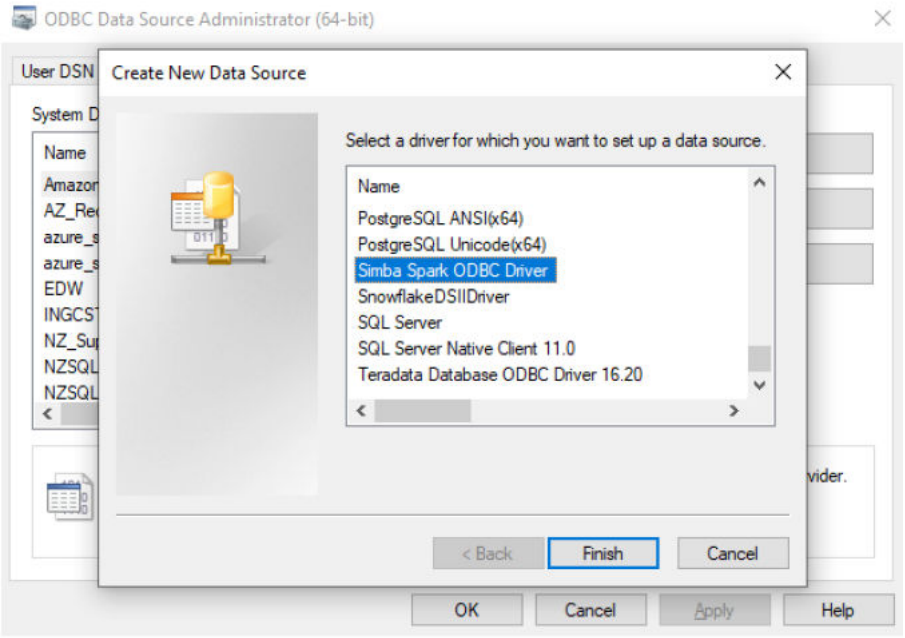
   You can download the Databricks ODBC driver for Windows here: https://docs.databricks.com/integrations/bi/jdbc-odbc-bi.html#odbc-windows.

2. Double-click on the dowloaded *.msi* file to install the driver.

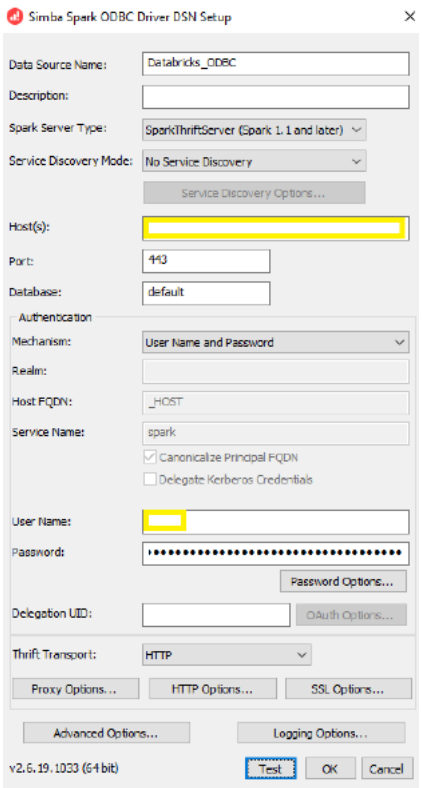
   The installation directory is: *C:\Program Files\Simba Spark ODBC Driver*.

3. From the **Start** menu, search for **ODBC Data Sources** to launch the ODBC Data Source Administrator.
4. Navigate to the **Drivers** tab to verify that the **Simba Spark ODBC Driver** is installed.

5. Go to the **User DSN** or **System DSN** tab and click **Add**.
6. Select **Simba Spark ODBC Driver** from the list of installed drivers as shown in the following image:



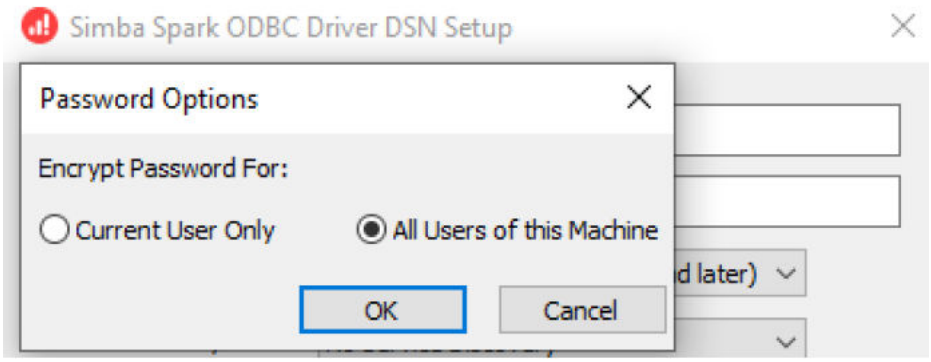
7. Click **Finish**.
8. Specify a **Data Source Name** and configure the mandatory ODBC configuration and connection parameters as shown in the following image:



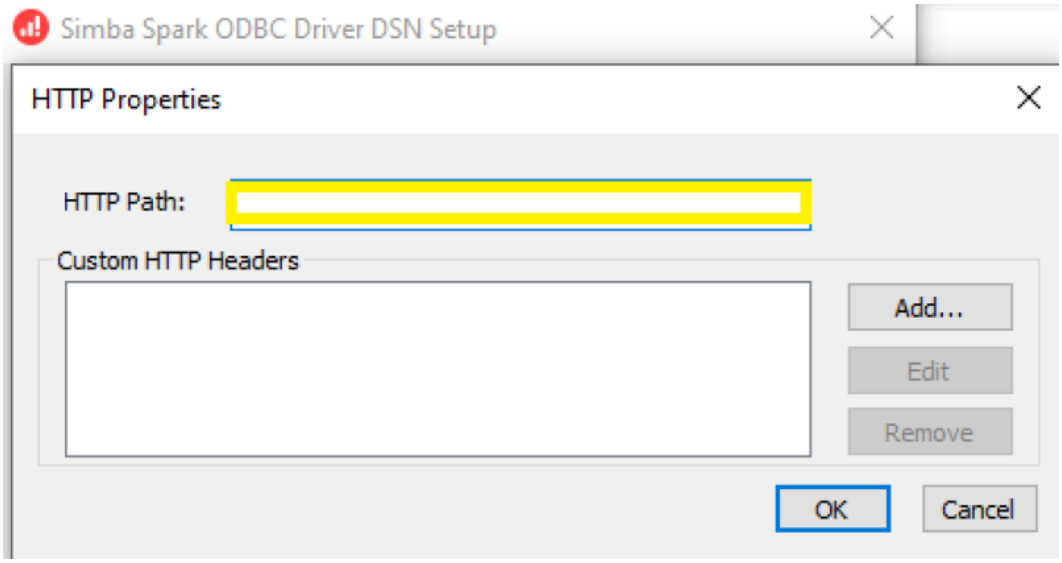
You can see details on the mandatory configuration and connection parameters for the ODBC driver here: https://docs.databricks.com/integrations/bi/jdbc-odbc-bi.html#odbc-parameters.

You can view the optimal ODBC driver capabilities here: https://docs.databricks.com/integrations/bi/jdbc-odbc-bi.html#odbc-driver-capabilities.

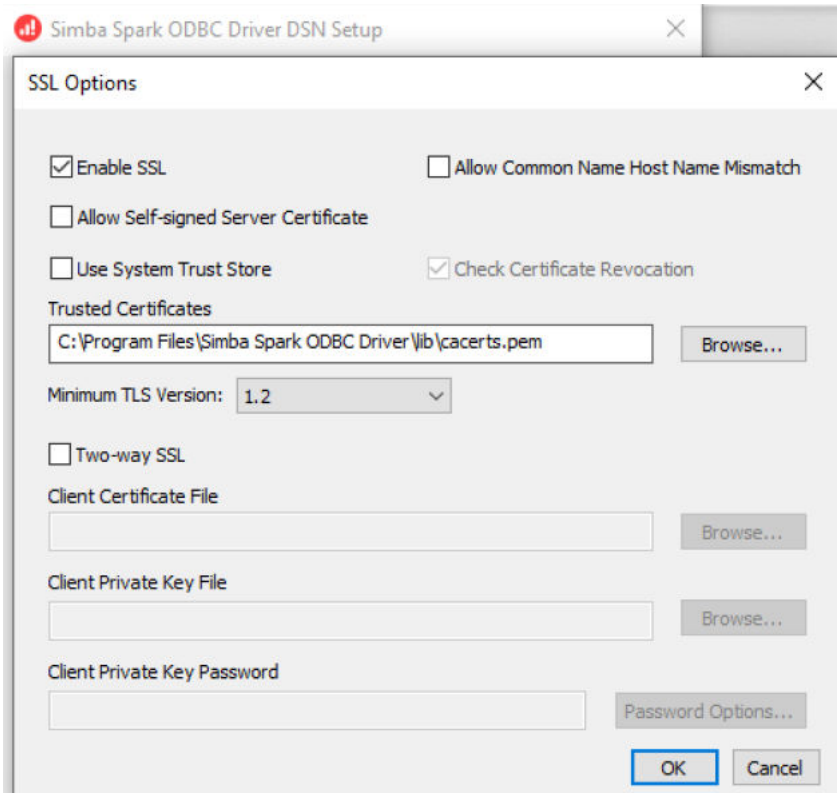
9. Click **Password Options** and choose to encrypt the password for all users of this machine as shown in the following image:



10. Click **HTTP Options** and type the HTTP path as highlighted in the following image:



11. Click **SSL Options** and specify the SSL properties as shown in the following image:



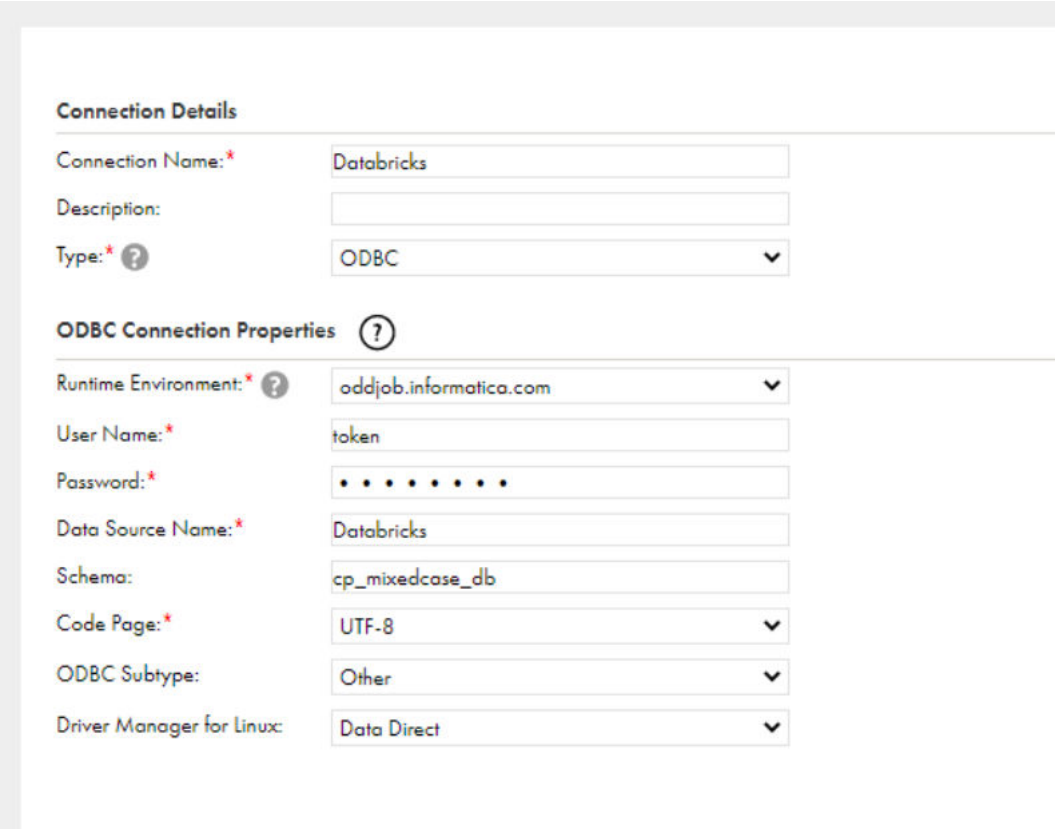
12. Click **Test** to test the data source configuration.
13. Click **OK** to create the DSN.

Note: You can retain the default values for the other options or you can tune the configuration according to the Databricks cluster.

Step 4. Create the connection in Administrator

Create a new connection of ODBC type on the **Connections** page in the Administrator service.

1. Go to the **Connections** page in the Administrator service.
2. Type the name of the connection.
3. Select the connection type as **ODBC**.
4. Specify the connection properties as shown in the following image:



5. Click **Test** to test the connection.

   If a database connection fails, contact the database administrator.
6. Click **Save** to save the connection.

Step 5. Create and run profiles

Create and run profiles on Databricks Delta tables using Azure Databricks with ODBC connection to view and analyze the content and structure of source objects.

1. In Data Profiling, click **New**.
2. In the **New** dialog box, click **Data Profiling Task**.

   The **Definition** tab for the profile appears.
3. On the **Definition** tab, enter the asset, source, and profile details. You can also choose columns and add a filter for the profile.
4. On the **Rules** tab, add one or more Data Quality assets as rules to the profile.
5. On the **Schedules** tab, optionally choose a runtime environment and schedule. You can also change the default email notification options and advanced options for the profile run as necessary.
6. Choose one of the following options to save and run the profile:
   - Click **Save** to save the profile.
   - Click **Run** to save and run the profile.

- Save the profile and choose a schedule on the **Schedules** tab to run the profile.

Run a profile using Azure Databricks with ODBC connection on Linux

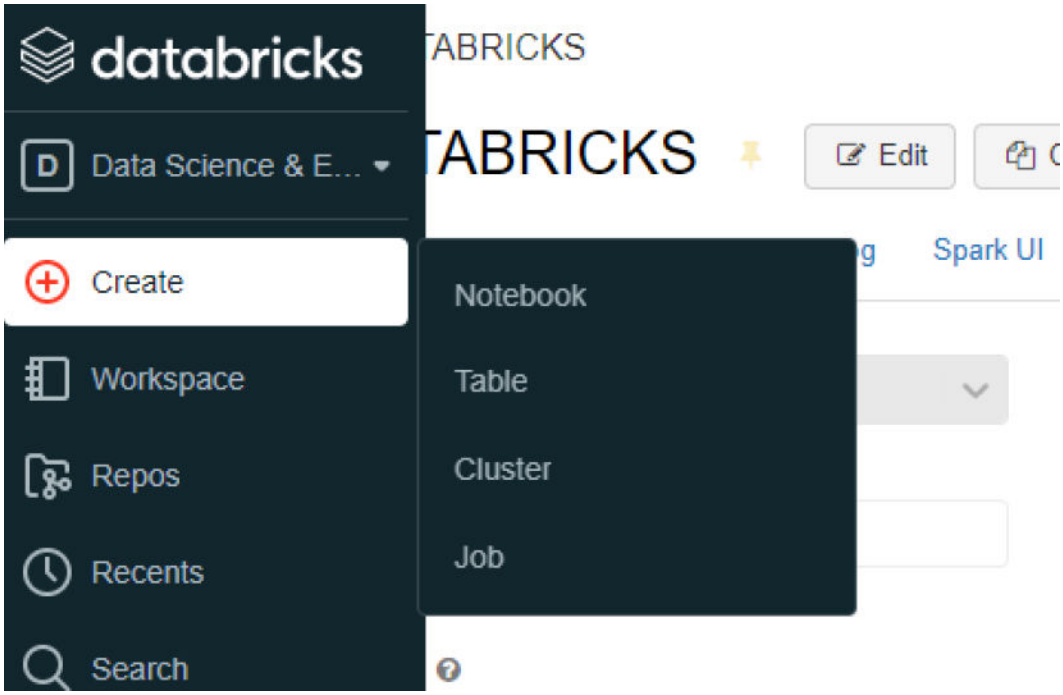
You can run a profile on Databricks Delta tables using Azure Databricks with ODBC connection on Linux.

Step 1. Create a cluster in Databricks

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2. Go to your workspace.
3. Click **Create** and choose **Cluster** as shown in the following image:



4. Enter a name for the cluster.

   You can continue with the default values for Worker type and Driver type.
5. Click **Create Cluster** and wait for the cluster to be up.

   Note: If you are using an existing cluster, make sure that the cluster is up and running.

Step 2. Retrieve the ODBC details

Retrieve the ODBC details from the Databricks cluster.

1. Go to the **JDBC/ODBC** tab of the Databricks cluster.
2. Retrieve the ODBC details including the sever host name, port, protocol, HTTP path, and the URL.

Step 3: Install and configure the ODBC driver for Linux

Set up a Data Source Name (DSN) configuration in Linux to connect the ODBC client application to Databricks.

1. Download the latest Databricks ODBC driver for Linux.

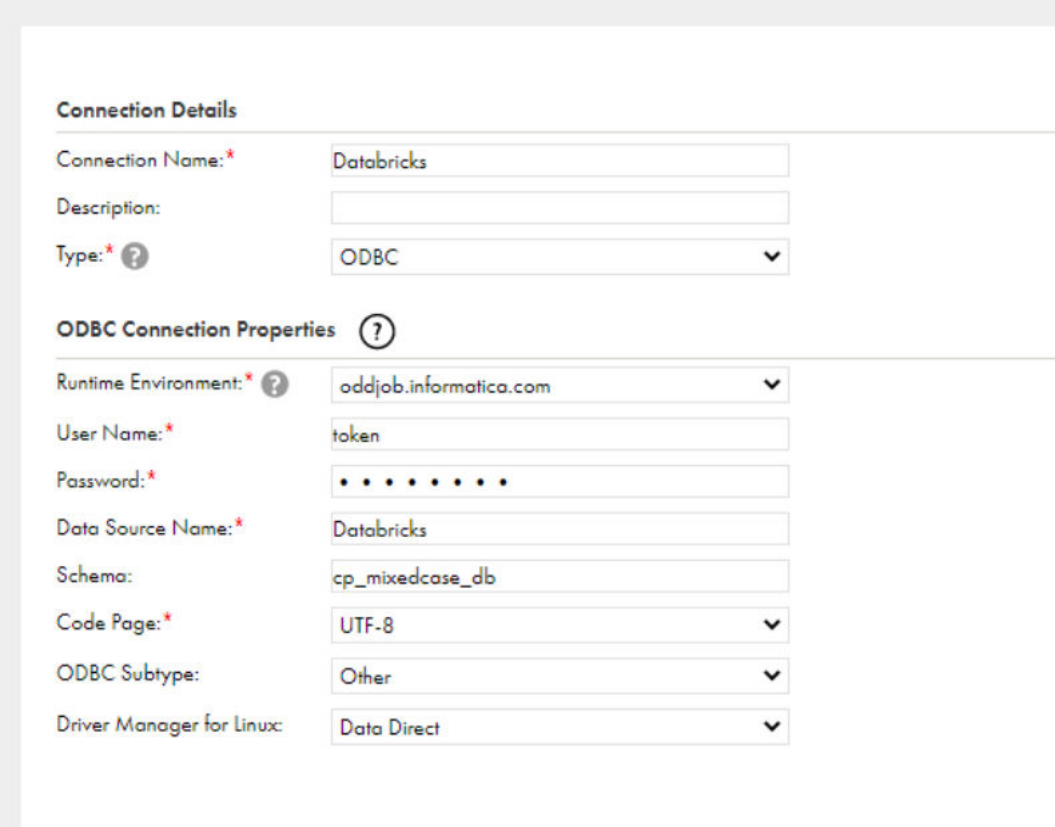
   You can download the Databricks ODBC driver for Linux here: https://docs.databricks.com/integrations/bi/jdbc-odbc-bi.html#odbc-linux.

   The Databricks ODBC driver is installed at: */opt/simba/spark*.
2. Create a DSN entry in the *odbc.ini* file with the ODBC details retrieved from the Databricks cluster.
3. Set the ODBC path in the *odbc.ini* file to */opt/simba/spark*.

Step 4. Create the connection in Administrator

Create a new connection of ODBC type on the **Connections** page in the Administrator service.

1. Go to the **Connections** page in the Administrator service.
2. Type the name of the connection.
3. Select the connection type as **ODBC**.
4. Specify the connection properties as shown in the following image:



5. Click **Test** to test the connection.

   If a database connection fails, contact the database administrator.
6. Click **Save** to save the connection.

Step 5. Create and run profiles

Create and run profiles on Databricks Delta tables using Azure Databricks with ODBC connection to view and analyze the content and structure of source objects.

1. In Data Profiling, click **New**.
2. In the **New** dialog box, click **Data Profiling Task**.

   The **Definition** tab for the profile appears.
3. On the **Definition** tab, enter the asset, source, and profile details. You can also choose columns and add a filter for the profile.
4. On the **Rules** tab, add one or more Data Quality assets as rules to the profile.
5. On the **Schedules** tab, optionally choose a runtime environment and schedule. You can also change the default email notification options and advanced options for the profile run as necessary.
6. Choose one of the following options to save and run the profile:
   - Click **Save** to save the profile.
   - Click **Run** to save and run the profile.
   - Save the profile and choose a schedule on the **Schedules** tab to run the profile.

Author

Nasmin N S